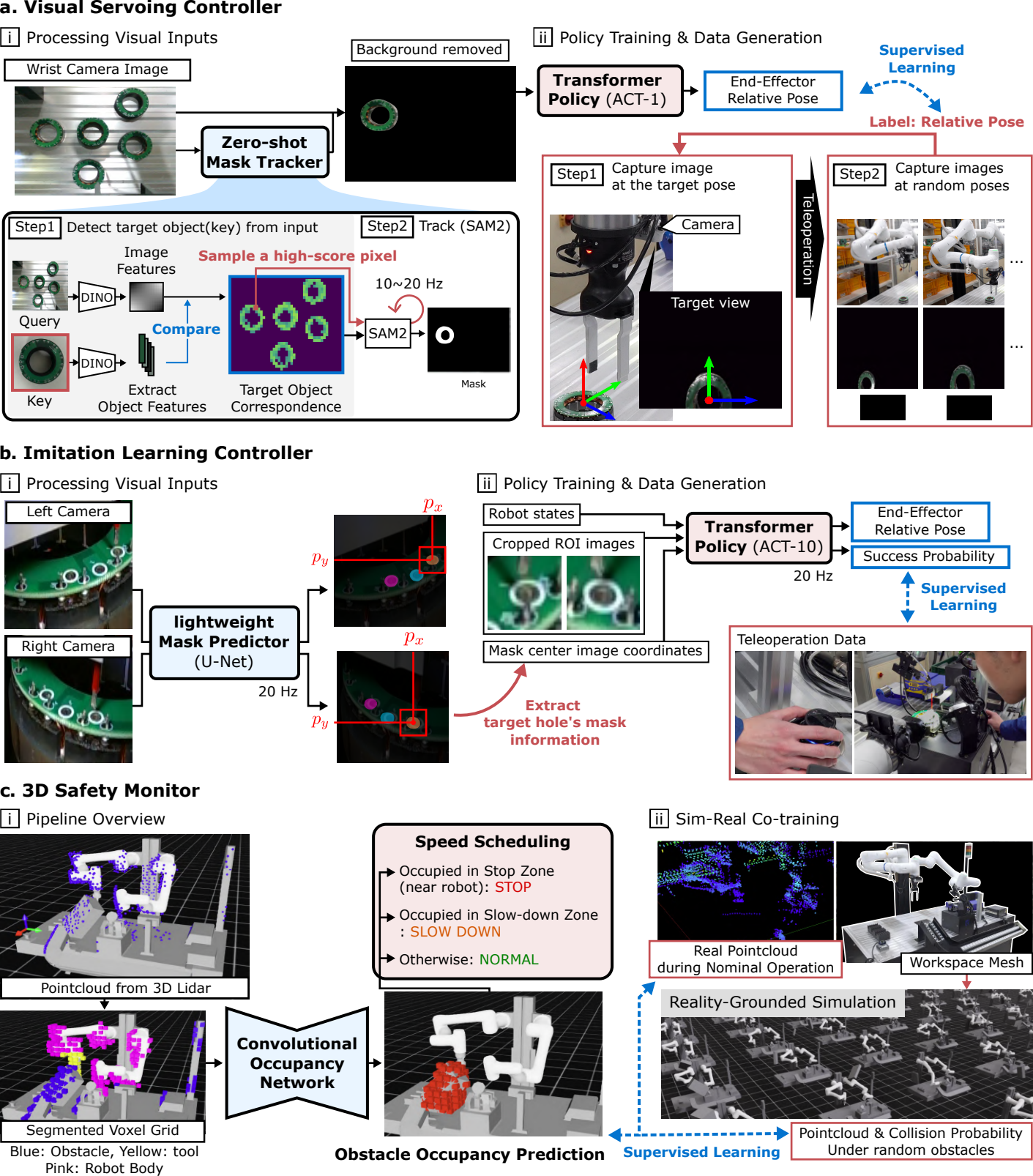

a. Visual Servoing Controller
i Processing Visual Inputs
Wrist Camera Image
Zero-shot Mask Tracker
Background removed
ii Policy Training & Data Generation
Transformer Policy (ACT-1)
End-Effector Relative Pose
Supervised Learning
Label: Relative Pose
Step1 Detect target object(key) from input
Image Features
Sample a high-score pixel
Query
DINO
Compare
Key
Extract Object Features
Target Object Correspondence
Step2 Track (SAM2)
10~20 Hz
SAM2
Mask
Step1 Capture image at the target pose
Camera
Target view
Teleoperation
Step2 Capture images at random poses
b. Imitation Learning Controller
i Processing Visual Inputs
Left Camera
Right Camera
lightweight Mask Predictor (U-Net)
20 Hz
p_x
p_y
ii Policy Training & Data Generation
Robot states
Cropped ROI images
Mask center image coordinates
Transformer Policy (ACT-10)
20 Hz
End-Effector Relative Pose
Success Probability
Supervised Learning
Extract target hole's mask information
Teleoperation Data
c. 3D Safety Monitor
i Pipeline Overview
Pointcloud from 3D Lidar
Segmented Voxel Grid
Blue: Obstacle, Yellow: tool
Pink: Robot Body
Convolutional Occupancy Network
Speed Scheduling
Occupied in Stop Zone (near robot): STOP
Occupied in Slow-down Zone : SLOW DOWN
Otherwise: NORMAL
Obstacle Occupancy Prediction
ii Sim-Real Co-training
Real Pointcloud during Nominal Operation
Workspace Mesh
Reality-Grounded Simulation
Supervised Learning
Pointcloud & Collision Probability Under random obstacles